



Adfinis

Plan. Innovatively. | **Build.** Sustainably. | **Run.** Resiliently.
Potential. Unlocked.



Sovereign Secrets Management with FOSS

How Adfinis Connects to Your Core Business





Leading Secrets Management Expert

Adfinis plans, builds, and runs mission-critical secrets management platforms for 7+ years!



80+ Successful
Projects



25+ Managed
Services



20+ Experienced
Specialists



Upstream
Maintainer



Steering Committee
Member



Community
Supporter





Secrets Management

Secrets Management

- › How are you managing your Secrets today?



Secrets Management

- › How are you managing your Secrets today?
 - › You mean Passwords?
 - › Shared Keepass
 - › BitWarden
 - › SealedSecrets in Git
 - › Ansible-Vault
 - › GitLab CI Secrets
 - › Plaintext in Git



Secrets Can Impose Huge Risks

- › Insecure secrets sharing across teams
- › Missing or only **sporadic secrets rotation**
- › Default or **hardcoded credentials**
- › **Secrets sprawl** across a multitude of services
- › **No audit trail** and traceability
- › Lack of policies and governance for secrets
- › **Improper integrations** for app workload





What is Secrets Management?

- › Practice to protect and integrate all secrets your workloads need to function
- › Focuses on machine, not human secrets
 - › Secrets management != Password manager
- › Secrets management solutions must be...
 - › super secure
 - › highly available
 - › flexible
 - › integratable
 - › developer-friendly



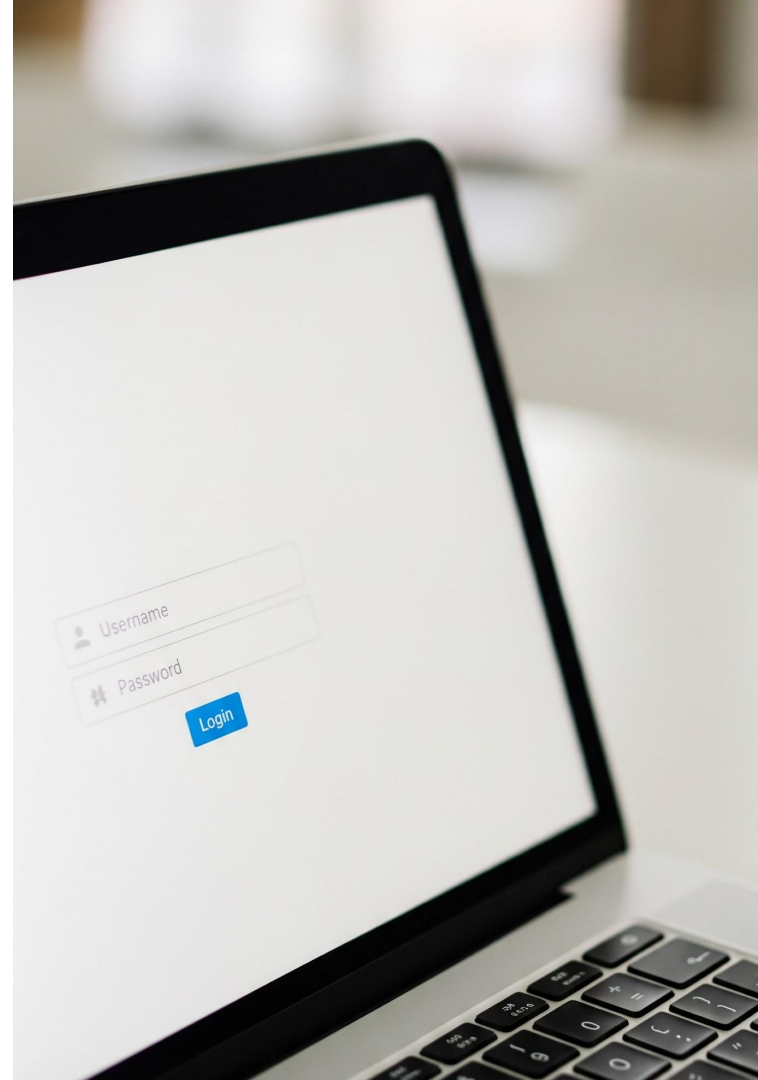
Human vs Machine Secrets

› Human Secrets

- › Manual: Created and managed by people
- › Long-lived: Persisted for extended periods
- › Threats: Prone to human mistakes
- › Accountability: Tied to specific users

› Machine Secrets

- › Automated: Generated programmatically
- › Short-lived: Temporary and on-demand
- › Threats: Lack of least privilege access
- › Accountability: Tied to specific machines



COMMUNITY
COLLABORATION
TRANSPARENCY

Free & Open Source Software



Why Should we Even Care About Open Source Software?

- › **Not because it's free to use!**
- › Collaboration & Innovation
 - › **Community** driven. Everyone is welcome.
- › Transparency & Trust
 - › Open code means **no secrets**. Easy inspection for security audit and building trust
- › Flexibility & Control
 - › Avoiding **vendor lock-in** by customizing the code for specific needs
- › Security
 - › **Vulnerabilities** are identified and fixed quicker
- › Sustainability
 - › No vendor means, the **software stays alive**, as long as there are people using and contributing to it



What Does it Mean to be Owned by a Foundation

- › Project is managed and overseen by a **non-profit organization** whose mission is to support and promote the software
- › The foundation usually handles various aspects of the project, including:
 - › Legal protection
 - › Financial stewardship
 - › Community governance
 - › Infrastructure support
 - › Advocacy and promotion
- › Good example: [The Linux Project](#)





The Linux Kernel Organization

The Linux Kernel Organization is a California Public Benefit Corporation established in 2002 to distribute the Linux kernel and other Open Source software to the public without charge. We are recognized by the IRS as a 501(c)3 private operating foundation.

- [IRS determination letter](#)
- [California determination letter](#)

2024-08-06

In [About](#).

The Linux Kernel Organization is managed by [The Linux Foundation](#), which provides full technical, financial and staffing support for running and maintaining the kernel.org infrastructure.



A close-up photograph of five steamed dumplings, likely xiaolongbuns, arranged on a white ceramic plate. The dumplings are plump and have a characteristic pleated top. They are resting on a dark wooden table. In the background, a wooden chair and a blurred interior space are visible, suggesting a restaurant or dining room setting. A blue rectangular overlay is positioned in the lower-left corner of the image.

OpenBao is the Answer



OpenBao



Search Ctrl+

 THE **LINUX** FOUNDATION PROJECTS

Manage, store, and distribute sensitive data with OpenBao

OpenBao is an open source, community-driven fork
of Vault managed by the Linux Foundation.

What is OpenBao?



- › OpenBao is modern open source secrets management solution
 - › Static and dynamic secrets
 - › Certificates and identities
 - › Key management
- › Ensures secrets for applications workloads remain safe
 - › Strict access control
 - › Encryption at rest
 - › Auditability
- › Governed and part of the Open Source Security Foundation
 - › Initially forked from HashiCorp Vault and API compatible so far

Visit openbao.org for more information



Secrets Engines

- › OpenBao has a variety of secrets engines
 - › **Key/Value** (most used, least complex)
 - › **Database**
 - › Cassandra
 - › InfluxDB
 - › MySQL/MariaDB
 - › PostgreSQL
 - › Valkey
 - › Custom
 - › **Kubernetes Secrets Engine** (ServiceAccount creation a gogo)
 - › **PKI**



Secrets Engines ... continued

- › OpenBao has even more secrets engines
 - › LDAP
 - › RabbitMQ
 - › **SSH**
 - › TOTP
 - › **Transit**
 - › Encryption as a Service



Authentication Methods

- › OpenBao has a variety of authentication methods that can be used
 - › **AppRole**
 - › **JWT/OIDC**
 - › Kerberos
 - › **Kubernetes** (basically JWT on steroids)
 - › **LDAP**
 - › Login MFA
 - › RADIUS
 - › **TLS Certificates**
 - › **Tokens**
 - › **Username & Password**



A photograph of a server room with a dumpling plugged into a network port. The background shows server racks with glowing green lights and tangled cables. A blue semi-transparent banner is in the lower-left corner.

Demo

Demo Instructions

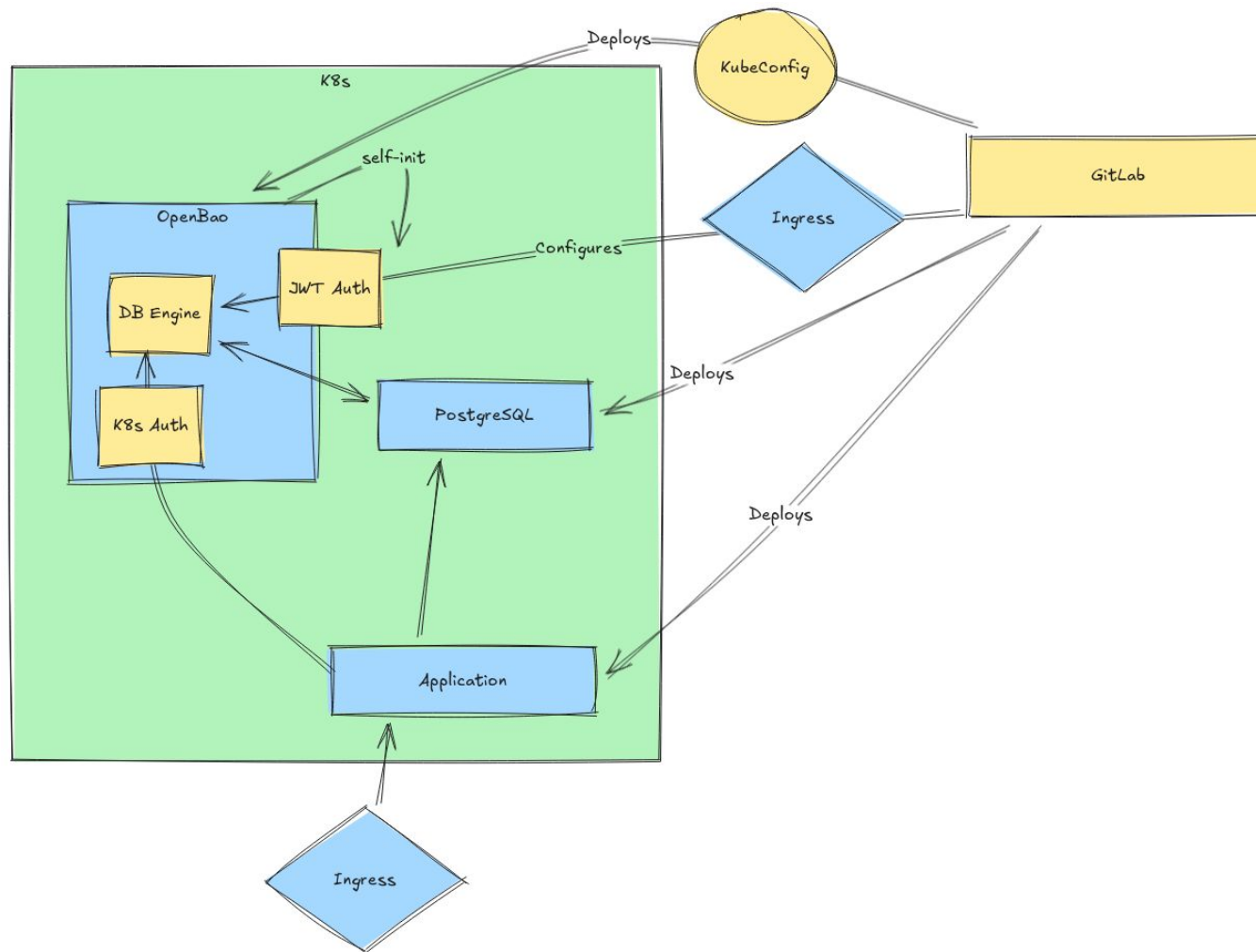
- › Deploy an application connecting to a database without having to manage any secrets at all over a GitLab Pipeline
- › We will deploy OpenBao on Kubernetes, and make it manage our secrets
- › Secret engines used:
 - › Database Secret Engine
- › Authentication Backends used:
 - › JWT Auth for GitLab
 - › Kubernetes Auth
- › OpenBao exclusive feature used:
 - › Self-initialization

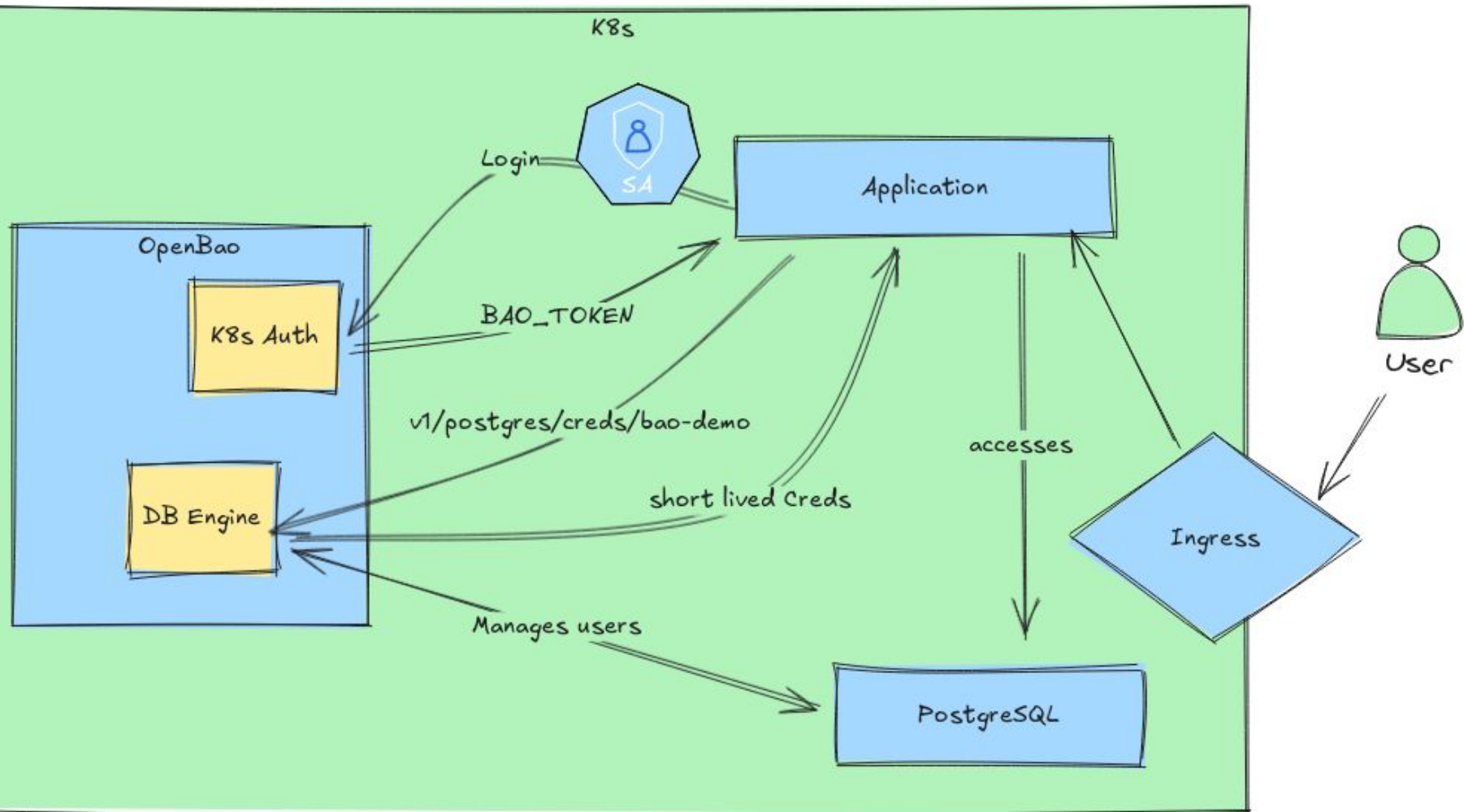


Demo Prerequisites

- › GitLab is already set up and can be used
- › Kubernetes is already set up and can be used (**k3s**)
- › \$KUBECONFIG is available to GitLab as CI/CD Variable
 - › But: [Using GitLab OIDC for K8s](#)
- › **No other secrets**







Demo Disclaimers

- › OpenBao **Static Seal** is not to be used in production!
- › Authenticate your GitLab instance with Kubernetes with short-lived secrets, not a long-lived kubeconfig
 - › [Using GitLab OIDC for K8s](#)





CONTRIBUTE & COLLABORATE

Centos® servers and environments
Fedora® processor and archive speedup/delay/delay
The various software volunteers

Contribute!

LINUX
FOUNDATION

Open Source
Initiative

How to Contribute?

- › <https://github.com/openbao/openbao>
- › <https://github.com/openbao/openbao-helm>
- › <https://openbao.org/docs/contributing>
- › Zulip Chat: [#openssf-openbao-discussion](#)
- › Community Call: every non-holiday Thursday, at 8AM PDT/11AM EDT/5PM CET on [Zoom](#)





`gitlab.com/adfinis/openbao-dem`

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**GOT
QUESTIONS?!?**



Let's Get in **Touch**



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